

Sainath Manda

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Summary

B.Tech final-year student with code already running in production under multiple projects. Merged PR's in C++ and cloud-native projects trusted by CNCF communities and scientific computing teams. Fluent in concurrent systems, GitHub workflows, and algorithmic problem-solving. Ready to join in September 2026 and contribute to Thoughtworks' client-facing engineering teams.

Education

CMR Engineering College Hyderabad	August 2022 – June 2026 (Expected)
B.Tech in Electronics and Communication Engineering	CGPA: 8.5 / 10.0
Sri Chaitanya Junior College Hyderabad	June 2019 – May 2021
Intermediate (High School)	Percentage: 94%

Technical Skills

Languages:	Go, C, C++, Java , Python, JavaScript, TypeScript, SQL
Backend & Infra:	Node.js, Express.js, REST APIs, Docker, Redis, RabbitMQ, Linux/Unix
Databases:	PostgreSQL, MySQL, SQL query design, schema modeling
Concepts:	Distributed systems, event-driven architecture, concurrency, microservices, TCP/IP networking

Open Source Contributions

HPX – C++ Standard Library for Parallelism and Concurrency	<i>STE AR Group Feb – May 2025</i>
- Implemented new parallel algorithm APIs (PR #7055) merged into the v2.0.0 master branch, extending the execution pipeline used by scientific computing teams and HPC researchers globally. - Resolved broken CMake build paths and refactored algorithmic headers (PR #7065), directly unblocking compilation for the v2.0.0 release milestone reviewed and approved by core maintainers.	
Meshery – Cloud Native Infrastructure Manager	<i>CNCF Project 2025 – Present</i>
- Fixed a critical icon rendering bug in the Registry action dialog's Relationships tab (PR #19884) by extending the NavItemDef TypeScript interface — restoring correct UI state across modal views. - Resolved a MUI Dialog z-index stacking context failure in the shared Modal wrapper (PR #19874), correctly forwarding typed sx props to eliminate component overlap in production UI.	

Projects

Rubik's Cube Solver – Advanced Algorithmic Engine	<i>C++, Graph Algorithms, Heuristics, State Space</i>
- Architected an optimal solver navigating a 4.3×10^{19} state space by implementing advanced CP graph algorithms (A* Search, IDA*) and custom Pattern Database heuristics in C++. - Optimized cube state representation using bit manipulation to minimize memory overhead, drastically pruning the search tree to achieve sub-second resolution of complex scrambles.	
LeetCode AI Student Bot – Code Mentor	<i>Go, React, C++, LLM APIs, WebSockets</i>
- Engineered a full-stack web platform with a React UI and a concurrent Go API backend, integrating external LLM APIs to stream real-time, context-aware DSA hints without spoiling solutions. - Built a sandboxed C++ execution engine communicating via gRPC to securely compile, run, and dry-test user code submissions against predefined edge-case test suites.	
Pebble – Fintech Auto-Savings Platform	<i>React, Golang, PostgreSQL, RabbitMQ, Redis</i>
- Developed an end-to-end full-stack application featuring a responsive React UI backed by a Golang microservices architecture processing 1,000+ concurrent rules/sec. - Built a real-time transaction dashboard utilizing WebSockets and an async RabbitMQ pipeline, integrating Redis caching to reduce database read latency by 40%. - Containerized both frontend clients and backend services via Docker Compose with Make automation, enabling single-command provisioning for the entire stack.	

Certified by

Indian CP Summer Camp: Algozenith – From the hands of ICPC World Finalist	July 2025
Advanced CPP Training: IIT Bombay	Oct 2024
Proficient in Problem Solving DSA: Hackerrank – Algorithms, Graphs, Linked list	July 2025

Achievements

- Selected for the **Indian CP Summer Camp** by Algozenith, ranking in the **top 50 out of 1,000+** competitive programmers nationwide.
- Consistently ranked as a **Top 50 Performer** across three consecutive advanced algorithmic mock contests during the CP Bootcamp.
- Solved **350+ coding problems** across Codeforces, LeetCode, and Algozenith, specializing in graph theory, dynamic programming, and systems-level problem patterns.